



Punyashlok Ahilyadevi Holkar Solapur University, Solapur

First Year B. Tech. (All Branches) Semester-I/II

BSC-01 ENGINEERING PHYSICS

(w.e.f. 2023-24)

Teaching Scheme

Theory - 3 Hrs/Week, 3 Credits

Laboratory - 2 Hrs/Week, 1 Credit

Examination Scheme

Theory - ESE- 70 Marks

ISE- 30 Marks

ICA - 25 Marks

- Course Objectives:**

BSC-01.01	To make students understand the categorization of solids based on band theory, ideas of Fermi level position in semiconductors.
BSC-01.02	To prepare students understand the different structures of solids and its characterizations.
BSC-01.03	To introduce students sound engineering through basic concepts of acoustics and ultrasonic.
BSC-01.04	To make students recall and discuss the concepts of wave optics via diffraction and polarization phenomenon.
BSC-01.05	To prepare students comprehend basics and working of LASER.
BSC-01.06	To prepare students understand the fundamentals of optical fiber and Nanoscience

- Course Outcomes:**

At the end of this course, student will be able to

BSC-01.1	Understand the concepts of semiconductors to implement in Engineering field.
BSC-01.2	Analyze the structural properties of elemental solids.
BSC-01.3	Apply concepts of acoustics and ultrasonic in various Engineering instruments.
BSC-01.4	Implement diffraction and polarization phenomenon to design advanced optical instruments.
BSC-01.5	Understand the fundamentals, construction and working of Laser in order to apply this Technology in Engineering fields.
BSC-01.6	Understand the basics of optical fiber and Nanoscience to implement in Engineering.

- **Course Curriculum:**

Section-I

Unit No. 01: Semiconductor Physics		Hours: 07		Marks: 16	
<i>Sr. No.</i>	<i>Subunit</i>	<i>Marks</i>	<i>Hours</i>	<i>Assessment</i>	<i>Bloom's Level</i>
1.1	Introduction to band theory of solids, Classification of solids, Fermi level (definition), Fermi-Dirac distribution function (introduction only)	4	3	Explanation, Definition, Classification	Remembering, Understanding
1.2	Fermi level in intrinsic and extrinsic semiconductors, effect of impurity concentration on Fermi level, Derivation for E_{Fin}	4	2	Explanation, Derivation	Remembering, Understanding
1.3	Hall Effect and applications	5	1	Derivation	Understanding
1.4	Numerical(s)	3	1	Numerical	Applying

Unit No. 02: Structure of Solids and its characterization		Hours: 08		Marks: 19	
<i>Sr. No.</i>	<i>Subunit</i>	<i>Marks</i>	<i>Hours</i>	<i>Assessment</i>	<i>Bloom's Level</i>
2.1	Introduction to crystal systems, characteristics of cubic unit cell: number of atoms per unit cell, atomic radius, co-ordination number	5	3	Explanation, Definition	Remembering, Understanding
2.2	Atomic packing factor, void space, density of crystal, Bragg's Law	3	2	Explanation, Describe	Remembering, Understanding
2.3	Symmetry elements (axis, center and plane)	5	1	Explanation, Describe	Remembering, Understanding
2.4	Miller indices, inter planner distance (by using Miller indices)		1	Derivation	Remembering, Understanding
2.5	Numerical(s)	6	1	Numerical	Applying

Unit No. 03: Sound Engineering		Hours: 06		Marks: 16	
<i>Sr. No.</i>	<i>Subunit</i>	<i>Marks</i>	<i>Hours</i>	<i>Assessment</i>	<i>Bloom's Level</i>
3.1	Acoustics: Introduction, reverberation, reverberation time, absorption coefficient (definition only), Sabine's formula, basic requirements for acoustically good hall	4	2	Explanation, State/Define	Remembering, Understanding, Applying
3.2	Factors affecting acoustics of auditorium and their remedies,	5	1	Explanation	Remembering, Understanding
3.3	Numerical(s)	3	1	Numerical	Applying
3.4	Ultrasonic: Introduction, piezoelectric effect and magnetostriction effect (introduction), Properties of ultrasonic waves, Detection methods of ultrasonics and its applications	4	2	Explanation, State	Remembering, Understanding

Section-II

Unit No. 04: Wave Optics		Hours: 08		Marks: 19	
<i>Sr. No.</i>	<i>Subunit</i>	<i>Marks</i>	<i>Hours</i>	<i>Assessment</i>	<i>Bloom's Level</i>
4.1	Diffraction: Introduction, resolving power, Rayleigh criterion, theory of diffraction grating.	4	3	Explanation, State/Define	Remembering, Understanding
4.2	Resolving power of diffraction grating	5*	1	Explanation	Understanding, Applying
4.3	Polarization: concept, optic axis, Malus law, positive and negative crystals, Optical activity, specific rotation	4	2	Explanation, State/Define	Remembering, Understanding
4.4	Laurent's half shade Polarimeter	5*	1	Explanation	Understanding, Applying
4.5	Numerical(s)	6	1	Numerical	Applying

*- indicates single count.

Unit No. 05: LASER		Hours: 06		Marks: 14	
<i>Sr. No.</i>	<i>Subunit</i>	<i>Marks</i>	<i>Hours</i>	<i>Assessment</i>	<i>Bloom's Level</i>
5.1	Interaction of radiation with matter-Stimulated absorption, spontaneous and stimulated emission, population inversion, pumping, metastable state, properties of laser	6	3	Explanation, State/Define	Remembering, Understanding
5.2	He-Ne gas laser: Construction, working on energy level	5	2	Explanation, Describe	Understanding, Applying
5.3	Holography (construction and reconstruction), applications of laser (science, engineering and medical)	3	1	Explanation	Remembering, Understanding

Unit No. 06: Optical Fiber and Nanoscience		Hours: 07		Marks: 18	
<i>Sr. No.</i>	<i>Subunit</i>	<i>Marks</i>	<i>Hours</i>	<i>Assessment</i>	<i>Bloom's Level</i>
6.1	Introduction, structure of optical fiber, basic principle of optical fiber (Total Internal Reflection)	4	2	Explanation, State/Define	Remembering, Understanding
6.2	Classification of optical fibers-step and graded index fibers, advantages of optical fibers over conducting wires		1	Explanation, Classification	Remembering, Understanding
6.3	Derivation for acceptance angle, acceptance cone and numerical aperture, fractional refractive index change	5	1	Derivation	Remembering, Understanding
6.4	Numerical(s)	6	1	Numerical	Applying
6.5	Nanotechnology: Introduction, carbon Nanotubes and its classification, applications of nanotechnology (electronics, energy, automobiles, space and defence, medical, cosmetics, environmental)	3	2	Explanation, State/Define	Remembering, Understanding

- **In Semester Evaluation (ISE):**

In Semester Evaluation (ISE) marks shall be based upon student's performance in three tests conducted & evaluated at institute level.

- **Internal Continuous Assessment (ICA):**

Internal Continuous Assessment (ICA) marks are calculated based on student's performance during laboratory sessions (minimum **eight** experiments out of below list).

1. Measurement of band gap energy of a semiconductor diode.
2. To determine Hall coefficient and charge carrier concentration by Hall Effect.
3. Study of Crystal structures.
4. Determination of velocity of sound using Kundt's tube apparatus.
5. Determination of wavelength of light by using diffraction grating.
6. Resolving power of a telescope/ diffraction grating.
7. Verification of inverse square law of intensity of light (Malus law).
8. To determine specific rotation by using Laurent's Half Shade Polarimeter.
9. Calculation of divergence of LASER beam.
10. Determination of wavelength of LASER using diffraction grating.
11. Determination of inter-planar distance 'd' using XRD pattern.
12. Study of Fiber optics.
13. To determine the Numerical Aperture of the optical fiber.

- **Text / Reference Books:**

1. Engineering Physics, R. K. Gaur and S.L. Gupta, Dhanapat Rai Publications
2. A Text Book of Engineering Physics, M.N. Avadhanulu, P.G. Kshirsagar, S. Chand and Company Ltd.
3. Modern Physics, B.L. Theraja, S. Chand and Company Ltd.
4. A Text Book of Optics, Subramanya and BrijLal, S. Chand and Company Ltd.
5. Nanotechnology: Principles and Practices, Sulabha K. Kulkarni, Capital Publishing Company
6. Nanoscience and Nanotechnology: Fundamentals to Frontiers –M.S. Ramachandra Rao, Shubra Singh, Wiley India Pvt Ltd, New Delhi
7. Engineering Physics, D.K. Bhattacharya, Poonam Tandon, Oxford University Press
8. Solid State Physics, S.O. Pillai, McGraw Hill Publications